ELS_Interface — Test Procedure

This procedure is used to **verify that the board works** after assembly, before installing it on the machine. It is written for a bench technician: every test states what to measure, how to measure it, which value is expected and leaves a space to note the result.

What you need: the assembled ELS_Interface board, a NUCLEO-G474RE, a 12 V bench power supply, a multimeter, two 1 kOhm resistors. An oscilloscope is recommended but not mandatory.

How to read it: the tests are numbered **T1** to **T18** and must be run **in order**. If a test does not pass, stop and find the cause before going on: the following tests assume the previous ones are correct.

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1. Prerequisites

Before powering anything, check the following.

#	What to check	Done
P1	Board **not powered**: resistance between +12 V and ground must be **above 10 kOhm**	
P2	Board not powered: resistance between **E5V** and ground must be **above 1 kOhm** (below 100 Ohm there is a short: do not power up)	
P3	Visual inspection: no solder bridge, no rotated component	
P4	**C1** (100 uF electrolytic) and **C5** (220 uF tantalum) mounted with the correct polarity	
P5	**F1** is the 1.1 A polyfuse and **F2** the 0.5 A one (not swapped)	
P6	The **Nucleo is not inserted** for tests T1 to T7	
P7	The power supply is set to **12 V with a 300 mA current limit**	
P8	Every instrument has its ground connected to a single point (common ground)	

2. The tests

T1 — Power-up with no load

Field	Content
What it verifies	that the board has no short circuits and that the converter starts
Procedure	1. Connect the supply to **J3** (pin 1 = +12 V, pin 2 = ground). · 2. Set the voltage to 12.0 V. · 3. Read the current on the supply display.
Expected result	current **below 50 mA**
Measured result	

T2 — Converter output voltage (E5V)

Field	Content
What it verifies	that converter U1 produces the correct 5 V
Procedure	1. Board powered at 12 V. · 2. Put the positive probe on **L1 pad 2** (or C5 pad 1). · 3. Put the negative probe on ground. · 4. Read the voltage.
Expected result	**5.00 V +/- 0.15 V**
Measured result	

If the value is wrong: check the solder joint of the U1 thermal pad, the R1/R2 divider and the L1 and D2 solder joints.

T3 — Voltage after the protection

Field	Content
What it verifies	that F1 and D1 do not introduce abnormal drops
Procedure	1. Board powered. · 2. Measure the voltage on **C1 pad 1**.

Field	Content
Expected result	**12.0 V** with a drop on F1 below 50 mV
Measured result	

T4 — Board rail (V_5V_SHIELD)

Field	Content
What it verifies	that diode D4 feeds the internal rail correctly
Procedure	1. Board powered at 12 V. · 2. Measure the voltage on **C12 pad 1**.
Expected result	**4.4 – 4.7 V** (the 5 V minus the diode drop)
Measured result	

T5 — Ripple on the converter output

Field	Content
What it verifies	that the supply is "clean" and does not disturb the microcontroller
Procedure	1. Connect the oscilloscope probe (10x, ground spring) to **L1 pad 2**. · 2. Enable the 20 MHz bandwidth limit. · 3. Read the peak-to-peak amplitude.
Expected result	**below 50 mV peak-to-peak**
Measured result	

T6 — Thermal check

Field	Content
What it verifies	that nothing gets too hot
Procedure	1. Leave the board powered for **5 minutes**. · 2. Measure the temperature of U1, L1 and D2 (contact thermometer or thermal camera).

Field	Content
Expected result	no component above **50 deg C**; U1 and L1 warm to the touch
Measured result	

T7 — Display branch

Field	Content
What it verifies	that the display 12 V arrives and that F2 has not tripped
Procedure	1. Board powered. · 2. Measure the voltage between **J7 pin 1** and ground.
Expected result	**12.0 V**, F2 not tripped
Measured result	

T8 — Drop across diode D4

Field	Content
What it verifies	that D4 conducts correctly
Procedure	1. Board powered at 12 V. · 2. Measure the voltage between the anode (E5V side) and the cathode (rail side) of D4.
Expected result	**0.35 – 0.45 V**
Measured result	

T9 — Diode D5 blocks the back-feed

Field	Content
What it verifies	**the most important safety check of the board**: that the 5 V cannot enter the computer USB port
Procedure	1. Power the board at 12 V (Nucleo not inserted). · 2. Insert a **1 Ohm shunt** in series with pin **CN10-8 (J2-8)**. · 3. Measure the voltage across the shunt and derive the current (I = V / 1 Ohm).
Expected result	current **below 10 uA** (practically zero)
Measured result	

This test must be repeated every time the supply circuit is modified.

T10 — Nucleo insertion

Field	Content
What it verifies	that the shield powers the Nucleo board correctly
Procedure	1. With the board **off**, insert the Nucleo checking pin 1. · 2. Set **JP5 on pins 5-6 (E5V)**. · 3. Power at 12 V. · 4. Measure the **3.3 V** on J1 pin 12.
Expected result	the Nucleo turns on; **3.28 – 3.32 V**
Measured result	

T11 — Green LED chain

Field	Content
What it verifies	that LED1 and resistor R19 are fitted and working

Field	Content
Procedure	1. Board **off**. · 2. Apply **3.3 V through a 1 kOhm resistor** on the **R19** pad on the J1-28 side. · 3. Look at the LED and measure the drop across R19.
Expected result	LED1 lights (dim); drop across R19 **0.32 – 0.42 V**
Measured result	

T12 — Red LED chain

Field	Content
What it verifies	that LED2 and R18 are fitted correctly
Procedure	1. Board off. · 2. Apply 3.3 V through 1 kOhm on the **R18** pad on the J1-30 side.
Expected result	LED2 lights; drop across R18 **0.7 – 1.5 V**
Measured result	

T13 — Optocoupler U3

Field	Content
What it verifies	that the first isolated output works
Procedure	1. Connect a test load (LED + 1 kOhm to an external +5 V) on **J8 pin 1**. · 2. Apply **5 V through 1 kOhm** on the **R12** pad. · 3. Measure the voltage between J8-1 and the rail.
Expected result	the load turns on; saturation voltage **below 0.4 V**
Measured result	

T14 — Optocouplers U4 and U5

Field	Content
What it verifies	that the second and third isolated outputs work independently
Procedure	Repeat test T13 on the **R13** (J8-2) and **R14** (J8-3) pads.
Expected result	same behaviour, independent outputs, saturation **< 0.4 V**
Measured result	

T15 — Opto outputs off

Field	Content
What it verifies	that the optos do not conduct when they are not driven
Procedure	1. Remove the test supply used in T13/T14. · 2. Measure the residual current in the load.
Expected result	load off; residual current **below 10 uA**
Measured result	

T16 — Supplies and buttons on J5

Field	Content
What it verifies	that the 5 V reaches the field connector and that the pull-up resistors are present
Procedure	1. Measure between **J5-1** and ground: it must give the rail (4.4–4.7 V). · 2. Measure between **J5-11** (emergency) and ground. · 3. Short J5-11 to ground and measure again. · 4. Repeat on J5-15, 16, 14, 13 with the buttons.
Expected result	J5-1 = rail; J5-11 = **3.3 V at rest** and **< 0.3 V** when shorted to ground; buttons 3.3 V at rest, 0 V pressed

Field	Content
Measured result	

T17 — Level shifter U2 levels

Field	Content
What it verifies	that the 3.3 V to 5 V level shifter works
Procedure	1. Measure **U2 pin 1** (3.3 V side) and **U2 pin 14** (5 V side). · 2. Apply 3.3 V on U2 pin 2 (input A1). · 3. Measure the corresponding B1 side.
Expected result	pin 1 = 3.30 V; pin 14 = 4.4–4.7 V; B1 = about 0.95 x VCCB = **4.3 – 4.5 V**
Measured result	

T18 — Ground continuity

Field	Content
What it verifies	that all grounds are really connected together
Procedure	1. Board off. · 2. With the multimeter in continuity mode, check between the ground pins of all connectors: J1 (8, 19, 20, 22), J2 (9, 20), J3-2, J5-17, J7 (7, 8), J8-5.
Expected result	resistance **below 0.5 Ohm** between any pair of ground pins
Measured result	

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3. Acceptance criteria

The board is considered **tested** only if all of these points are satisfied.

#	Criterion	Tolerance	Test
1	E5V voltage	5.00 V +/- 0.15 V	T2
2	Internal rail voltage	4.4 – 4.7 V	T4
3	Ripple on the converter output	< 50 mV peak-to-peak	T5
4	**No back-feed into the USB port**	< 10 uA	T9
5	Drop across diode D4	0.35 – 0.45 V	T8
6	Temperature after 5 minutes	no component above 50 deg C	T6
7	Saturation of the three isolated outputs	< 0.4 V each	T13, T14
8	Residual current with the optos off	< 10 uA	T15
9	Ground continuity	< 0.5 Ohm	T18

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4. Registration sheet

Fill in this table during the test.

Test	Description	Measured value	Result (OK / KO)
T1	Power-up with no load		
T2	E5V voltage		
T3	Voltage after the protection		
T4	Internal rail		
T5	Ripple		
T6	Thermal check		
T7	Display branch		
T8	Drop across D4		
T9	Back-feed blocked		

Test	Description	Measured value	Result (OK / KO)
T10	Nucleo insertion		
T11	Green LED		
T12	Red LED		
T13	Opto U3		
T14	Opto U4 and U5		
T15	Optos off		
T16	Supplies and buttons J5		
T17	Shifter U2 levels		
T18	Ground continuity		

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5. Final outcome

Test date	
Technician	
Board (serial number or revision)	
Tests passed	_____ / 18
Acceptance criteria met	yes / no
Anomalies found	
Signature	

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6. Electrical design verification

Before the physical test, the design was checked with the **`kicad-mcp-pro`** MCP server (KiCad 10.0.6) and with `kicad-cli` :

Check	Result
ERC — electrical rules check of the schematic	**0 violations**
DRC — design rules check of the PCB	**0 active violations**, **0 unconnected items**
Netlist	exported correctly, **56 components**
Schematic / PCB parity	**56 = 56**, no missing component
3D models	**56 of 56** resolved

> **Note:** this table covers the physical board. Until a bench test is run, the measured values stay empty.